

False Face Must Hide What the False Heart Doth Know:

The Literary Face in the Age of AI

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ABSTRACT

This position paper examines a range of technological and algorithmic issues through the lens of literary and cultural theory about the face, in particular the ways in which emotion, expression and gesture are conveyed in literary texts. In doing this, it canvasses some of the primary aims and theoretical contexts of a new ARC project, *Literature and the Face: A Critical History*, which draws on literary, philosophical and visual traditions to generate comprehensive new knowledge about changing textual discourse about the face.

KEYWORDS

face, Literature, emotion, affect-detection-software, identity

ACM Reference Format:

Tyne D. Sumner. 2021. False Face Must Hide What the False Heart Doth Know:: The Literary Face in the Age of AI. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3460418.3479339>

1 FACIAL RECOGNITION TODAY

As facial recognition technologies continue to dominate discussion in fields ranging from Computing and Information Systems and surveillance studies through to psychology and urban design, attention has now turned to the use of automated emotion detection software (or affect recognition software) as part of the core infrastructure of many platforms around the world. Whereas facial recognition refers to the identification of individuals' faces for authentication purposes, emotion detection attempts to identify, categorize, and analyze human emotions by collecting information about the external appearance of any face. Automated affect-recognition systems are now being deployed in a wide range of settings ranging from hiring processes and security screenings through to supermarket advertising and dating websites.

The core driver of emotion-detection software is the gargantuan volume of data on the external images of human facial expressions that derive from both public and private images shared on platforms

including Facebook, Instagram, TikTok, Snapchat and Flickr. Similarly, mega technology companies such as IBM, Amazon, Google, and Microsoft are also employing emotion-detection software to attempt identify common emotions such as fear, sadness, and surprise. The highly lucrative and rapidly evolving field of automated affect-detection has prompted the need to critically consider the long history of emotions in the context of advancements in Artificial Intelligence that claim to be able to read a person's inner state from their outward appearance. This is particularly urgent in the case of predictive policing strategies, which use databases containing images of faces to train algorithms to detect potential criminal suspects.

The ways in which affect-recognition systems are trained on the founding assumption that there is a relatively small set of universal human emotions, each of which can be displayed on the external façade of the human face, also necessitates an urgent revisiting of the physiognomic and psychological research that underpins affect-detection's governing taxonomies. Such an approach, it could be argued, would benefit from a recentering of the history of emotions as developed in the Humanities and Social Sciences as well as a consideration of the ways that human experience and emotion have been articulated and critically analyzed in other fields such as art and literature for centuries.

By examining these pressing algorithmic issues through the lens of literature and cultural theory about the face, this paper proposes an innovative interdisciplinary approach to the study of affect-recognition technology. This approach forms part of a new Australian Research Council (ARC) Discovery Project, *Literature and the Face: A Critical History*, based at the University of Melbourne and the University of Geneva, the aim of which is to draw on literary, philosophical, and visual traditions to generate comprehensive new knowledge about changing textual discourse about the face. This project also seeks to trace the historical trajectory of several key facial 'tropes' across physiognomic and medical contexts through to their appearance in well-known literary texts, to consider the ways in which human emotions are understood not purely through objective visual signs but also via discursive rhetorical, semantic and textual modes.

2 READING THE FACE

In a 2019 study in *Psychological Science in the Public Interest*, researchers suggested "an urgent need for research that examines how people actually move their faces to express emotions and other social information in the variety of contexts that make up everyday life, as well as careful study of the mechanisms by which people perceive instances of emotion in one another." [1] This research signaled the need for new and diverse approaches to understanding

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UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA

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ACM ISBN 978-1-4503-8461-2/21/09...\$15.00

<https://doi.org/10.1145/3460418.3479339>

and critiquing the rise of facial-recognition technologies, affect detection tools and other similar digital technologies. More recently, in *Atlas of AI* (2021), Kate Crawford canvasses the precursor to automated affect recognition, beginning with the American psychologist Paul Ekman's suppositions about human emotions in the 1960s, alongside the current and burgeoning use of AI emotion detection today, noting that there are "powerful institutional and corporate investments in the validity of Ekman's theories and methodologies." [2] Other social research about the human face has come from the various branches of neuroscience that locate the cognitive and perceptual mechanisms of facial recognition and the developmental phases of understanding and interpreting facial expression as evidenced in works such as Bruce and Young's *Face Perception* (2012), *The Oxford Handbook of Face Perception* (2011), and Freedberg's *Feelings on Faces: From Physiognomics to Neuroscience* (2014). [3] In the computer sciences, computer vision and machine learning are increasingly employed in attempts to 'read' the face by deploying algorithms trained on Ekman's six categories of facial expression: happiness, sadness, surprise, fear, anger, and disgust.

The limitations of this approach—the assumption that internal emotions are automatically displayed externally via the face—are now apparent. Studies from a range of fields have emerged emphasizing the subjective experience of emotions and the facial movements that express them as well as the highly context dependent nature of expressive facial signals as mediated through such things as back-channeling, innuendo, cognitive load, interjection etc. In a 2016 paper, for example, Gunes and Hung argue that "the six categories of emotions have no use for the majority of everyday applications." Acknowledging this, they argue, necessitates moving into "uncomfortable territory where we have to ask ourselves the more fundamental questions like: what is the contemporary definition of emotion in this technologically-driven fast-changing world that is very different from that of Darwin's? How are these emotions represented in facial expressions?" [4]

3 FALSE FACE MUST HIDE WHAT THE FALSE HEART DOTH KNOW

The most direct way to think of the face is in visual terms, as imagined and represented through portraiture, photography, sculpture, film, and other visual media. Faces are central to the cultural, political, and social history of the visual arts, as discussed in the work of Mulvey (1975) and Berger (2002). [5] However, verbal and textual representations of the face also present highly complex and productive sites of cognitive and emotive meaning, even though they have been relatively sidelined in the research. Scholars such as Baumbach (2008) and Frow (2014) have addressed the face through individual texts and key narrative moments of facial interpretation, [6] while others have analyzed a single facial feature or trope across multiple disciplines or in the context of a single era, such as Trumble's *A Brief History of the Smile* (2004) and Jones' *The Smile Revolution in Eighteenth-Century Paris* (2017).

Literary texts present a crucial site for studying and analyzing emotional expression through the face, as well as instances of individual subjectivity and gendered, racial, and cultural identity. This is possible because of the ways that literature dramatizes, enacts,

stages, and encourages cognitive and emotional acts of comprehension and empathy through multiple modes of 'face reading'; between characters, narrators, and readers. In this way, literature offers a critical and timely challenge to the assumptions and algorithmic design principles of new and emerging affect-recognition technology because of its complex and diverse, rather than categorical, treatment of the human face. Texts such as poems, plays and novels also frequently establish the face as a site of contestation, resistance, innuendo, and duplicity. William Shakespeare's characters frequently display deceptive facial traits, which contradict their intentions and inner emotive states. In Act 1, Scene 7 of *Macbeth*, for example, when Macbeth resolves to murder King Duncan and take the throne of Scotland for himself, his speech hints at the duplicitous and misleading capacity of the face when he says: "Away, and mock the time with fairest show: / False face must hide what the false heart doth know."

As this well-known scene from Shakespeare suggests, the representation and interpretation of the face has been at the centre of the experience of literary reading for centuries. As readers, we engage in a complex discursive and interpretive act when we understand the ways in which characters in literary texts 'read' each other's faces by describing, reacting, pathologizing, interrogating and misreading them. The vehicle for enacting this process is literary language, through which visual images of the face (and, correspondingly, the emotions they transmit and evoke in readers) are registered through precise rhetorical and semantic maneuvers of language. Thus, through literary texts, the visual impression of the face is rendered via a complex verbal network of sense and meaning-making that is implicated in social and cultural contexts.

4 ASPIRING FOR AFFECT

An attention to the different textual and rhetorical forms and techniques by which the face is mediated affords a new interdisciplinary methodological framework for considering the motivations for and consequences of contemporary emotion-detection software. This is in large part because literature is central to the history of emotions and offers new ways of conceptualizing the present through the past. As Stephanie Trigg has written in relation to the complex facial tropes in the novels of Jane Austen, although literary language "often avoids the direct naming of emotions, it deploys a range of metaphorical expressions and structures to build rich and layered emotional landscapes." [7] Unlike algorithmic attempts to categorize and analyze the face's external signs for inner emotive state, literature presents a complex and historically situated archive, "not just for the expression of emotions, but also for our comprehension of the social exchange of emotions on several levels." [8] By attending to the ways that descriptions of faces and intersubjective exchanges in literary texts articulate and interrogate such things as empathy, feeling, subjectivity, identity, authenticity and concealment, this framework allows us to think about the human emotions and experiences that algorithmic processes are incapable of capturing, let alone analyzing. Doing this necessitates reconstructing a series of historical and evolving regimes of interpretation that have shaped, and continue to shape, the way faces are coded and decoded.

ACKNOWLEDGMENTS

This paper is supported by the Australian Research Council (ARC) Discovery Project, Literature and the Face: A Critical History in the School of Culture and Communication at the University of Melbourne and the University of Geneva. (face.hypotheses.org)

REFERENCES

- [1] Barrett, Lisa. et. al. "Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movement." *Psychological Science in the Public Interest* 20 (2019): 1-68.
- [2] Crawford, Kate. *Atlas of AI*. New Haven: Yale University Press (2021): 75
- [3] Bruce, V. and A. Young, *Face Perception*. Psychology Press (2012); Rhodes, G. A. et. al., eds. *The Oxford Handbook of Face Perception*. Oxford UP (2011); Freedberg, D. 'Feelings on Faces: From Physiognomics to Neuroscience,' in *Rethinking Emotion*, eds. R. Campe and J. Weber. De Gruyter (2014): 289-324.
- [4] Gunes H. and Hung, H. 'Is Automatic Facial Expression Recognition of Emotions Coming to a Dead End?' *Image and Vision Computing* 55.1 (2016): 6-8.
- [5] Mulvey, L. 'Visual Pleasure and Narrative Cinema.' *Screen* 16.3 (1975): 6-18; Berger, H. *Fictions of the Pose*. Stanford University Press (2000).
- [6] Baumbach, S. *Shakespeare and the Art of Physiognomy*. Humanities E-Books (2008); Frow, J. *Character and Person*. Oxford UP (2014).
- [7] Trigg, S. 'Faces that speak: A little emotion machine in the novels of Jane Austen.' In *Spaces for Feeling: Emotions and Sociabilities in Britain, 1650-1850*. Routledge (2015): 9.
- [8] *Ibid.*